

Donovan Reynolds

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Education

University of Central Florida

Master of Science in Computer Engineering. GPA(4.0/4.0)

Orlando, FL

University of Central Florida

Bachelor of Science in Computer Engineering. GPA(3.9/4.0)

Orlando, FL

Skills

Programming Languages: C, C++, Python, x86 Assembly, Java, Bash

Hardware & Microcontrollers: MSP430, ATmega (AVR Architecture), Jetson Nano, Raspberry Pi Pico, Raspberry Pi 4, JTAG Debugging, Sensor Integration

Development Tools: Linux/Unix, Git, Antlr, Code Composer Studio, openCV

Personal Projects

B.I.R.D.S.

- Designed an algorithm to aid in training birds to pick up cigarettes
- Engineered real-time sensor fusion software in Embedded C++ to process input data through the system
- Trained and deployed a custom Computer Vision model using openCV on Jetson Nano for real-time target detection
- Engineered a Python script to interface the image model and send data to an embedded device

Arduscript

- Designed a custom compiler in Python
 - Created a custom scripting language that is parsed by the Antlr toolchain.
 - Compiles into Arduino IDE compatible code to be flashed to any compatible system
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Experience

UCF

Orlando, FL

Graduate Instructor Systems Software

August 2025 - June 2026

- Led lab sections covering Linux system calls, shell development, and compiler design for undergraduate students.
- Guided students through the implementation of a custom shell and basic compiler projects using C/C++.
- Mentored students in debugging, code optimization, and software development best practices in Linux environments.
- Reinforced core systems programming concepts through hands-on instruction and technical support.

UCF

Orlando, FL

Undergraduate TA Embedded Systems

August 2023 - August 2024

- Facilitated lab sessions teaching embedded systems concepts using the MSP430 microcontroller platform.
- Supported students in developing and debugging C programs using JTAG and Code Composer Studio.
- Interpreted technical documentation to troubleshoot hardware/software issues and improve lab efficiency.
- Provided one-on-one technical guidance to strengthen student understanding of low-level programming concepts.

Academic Community Enrichment

Mims, FL

High School Math Teacher

August 2023 - May 2025

- Designed and taught introductory Python programming curriculum for high school students.
- Delivered instruction in mathematics ranging from Pre-Algebra through Pre-Calculus to diverse learning levels.
- Adapted teaching strategies to improve student comprehension, engagement, and academic performance.
- Built strong classroom communication and problem-solving skills through individualized student support.

Mathnasium of Lake Mary

Lake Mary, FL

Lead Instructor

Feb 2020 - August 2023

- Led and trained a team of instructors to effectively teach students across a wide range of math skill levels.
- Coordinated staff development and instructional strategies to improve student learning outcomes.
- Resolved scheduling, instructional, and student engagement challenges in a fast-paced tutoring environment.
- Fostered a collaborative team culture focused on high-quality instruction and customer satisfaction.